

Enhanced Private Pilot Certification Course**LESSON 124 (0.5 GROUND and 1.0 DUAL)****Lesson Objective**

The purpose of this lesson is to introduce the student to short and soft-field takeoffs and landings. The student will gain an understanding of why and how these operations are conducted.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed
 - Chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
 - Chapter 05 of the Pilot's Handbook of Aeronautical Knowledge - Aerodynamics of Flight
 - Wingtip Vortices.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 07 - Ground Reference Maneuvers.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 14 - Airport Operations.

Lesson Content (Ground)**Explanation of Short-Field and Soft-Field Takeoffs & Landing****Procedures and Techniques**

- Short field takeoff:
 - Purpose:
 - Short runway or obstacle at the end of the runway.
 - "To depart from such an area safely, the pilot must exercise positive and precise control of airplane attitude and airspeed so that takeoff and climb performance result in the shortest ground roll and the steepest angle of climb" (FAA, 2021).

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- In order to achieve that, V_X must be maintained until clear of the obstacle.
- Re-introduce V_X and explain if needed.
- Flaps should be set to 25 degrees. This increases the camber of the wing and therefore increases the amount of lift produced.
- Takeoff roll:
 - When entering the runway, use the maximum amount of runway available.
 - Once lined up with the centerline, stop on the runway, hold the brakes, and apply full power.
 - Release the brakes once the engine is at full static RPM.
 - Crosswind correction must be maintained.
 - Rotation is at 55 knots.
- Departure:
 - Once off of the runway, pitch for 63 knots until clear of the obstacle.
 - Emphasize to the student to not chase the airspeed. Instead, set a pitch attitude that will get the airplane to a particular airspeed.
 - Announce clear of obstacle.
 - Announce positive rate and retract the flaps to 10 degrees.
 - Announce positive rate and airspeed acceleration and retract the flaps to 0 degrees.
 - Continue the climb at 79 knots.
- Soft field takeoff:
 - Purpose:
 - "Takeoffs and climbs from soft fields require the use of operational techniques for getting the airplane airborne as quickly as possible to eliminate the drag caused by tall grass, soft sand, mud, and snow and may require climbing over an obstacle" (FAA, 2021).
 - "The technique makes judicious use of ground effect to reduce landing gear drag and requires an understanding of the airplane's slow speed characteristics and responses" (FAA, 2021).
 - Flaps should be set to 25 degrees. This increases the camber of the wing and therefore increases the amount of lift produced.
 - Takeoff roll:
 - When entering the runway, pull the yoke all the way aft to reduce the weight on the front wheel.
 - Do not stop on soft field environments.
 - Minimal braking is to be used, as this can result in the nose gear getting stuck in the grass, mud, sand, etc.
 - As full power is applied, full aft stabilator must be kept.
 - As the nose wheel gets off the runway, release the backpressure slightly to help the main gear get off the runway.

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- Once the airplane is fully off the runway, apply forward stabilator pressure to stay in ground effect until V_x or V_Y (depending if there is an obstacle).
- Crosswind correction must be maintained.
- Departure:
 - After rotation, if there is no obstacle, climb out at V_Y .
 - If there is an obstacle, climb out at V_x .
 - Announce positive rate/clear of obstacles and retract the flaps to 10 degrees.
 - Announce positive rate and airspeed acceleration and retract the flaps to 0 degrees.
 - Continue the climb at 79 knots.
- Short field landing:
 - Purpose:
 - “Short-field approaches and landings require the use of procedures for approaches and landings at fields with a relatively short landing area or where an approach is made over obstacles that limit the available landing area” (FAA, 2021).
 - Precise and positive control of the rate of descent and airspeed is required in order to achieve a successful short field landing.
 - The landing must result in little floating in order to stop the airplane in the shortest distance.
 - Approach and landing:
 - A higher than normal approach is required in order to clear the obstacle on the departure end of the runway.
 - Adjust power to maintain a steeper than normal angle of descent.
 - TRIM!
 - Short field approach speed: 63 knots (1400-1500 RPM).
 - Short final speed: 55 knots ($1.2V_{SO}$).
 - At the round out, continue the flare to touchdown on the main wheels first in approximately the pitch attitude that will result in a power-off stall just as the power is reduced to idle.
 - Landing roll:
 - Upon touchdown, hold this pitch attitude as long as the stabilator remains effective to provide aerodynamic braking, as appropriate.
 - Brakes should be applied evenly and firmly to minimize the after landing roll without skidding.
- Soft field landing:
 - Purpose:
 - “Landing on fields that are rough or have soft surfaces, such as snow, sand, mud, or tall grass, require unique procedures. When landing on such surfaces, the objective is to touch down as smooth as possible and at the slowest possible landing speed” (FAA, 2021).

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- “A pilot must control the airplane in a manner that the wings support the weight of the airplane as long as practical to minimize drag and stresses imposed on the landing gear by the rough or soft surface” (FAA, 2021).
- The airplane must be held into ground effect in order to touch down softly at minimum speed.
- Approach and landing:
 - The approach is identical to a normal approach.
 - At the round out, reduce the power to idle and stay in ground effect.
 - Begin a flare once the runway seems to be coming up to the airplane.
 - If necessary, a slight addition of power during touchdown will aid in softening the touchdown
- Landing roll:
 - Hold sufficient back-stabilator pressure to keep the nosewheel off the ground until it can no longer aerodynamically be held off the field surface.
 - Use minimal braking.

Steep Turns

- Purpose:
 - Steep turns consist of single to multiple 360° and 720° turns, in either or both directions, using a bank angle between 45° and 60°.
 - To develop a pilot’s skill in flight control smoothness and coordination, an awareness of the airplane’s orientation to outside references, division of attention between flight control applications, and the constant need to scan for hazards and other traffic in the area.
- Talk about the following fundamental concepts:
 - Higher G-forces.
 - The airplane’s inherent overbanking tendency.
 - Significant loss of the vertical component of lift when the wings are steeply banked
 - Substantial pitch control pressures.
 - The need for increased additional power to maintain altitude at a constant airspeed during the turn.
- Procedure:
 - Begin the maneuver at 90 KIAS.
 - Select prominent outside visual reference points.
 - Smoothly roll into a bank angle of 45 degrees.
 - Maintain altitude and airspeed by adjusting the pitch, bank, and power as necessary.
 - The rollout from the turns should be timed, approximately at ½ the bank angle, so that the wings reach level exactly on the entry heading.
 - After completing a 360-degree turn, roll wings level and prepare to repeat the maneuver in the opposite direction.

Wake Turbulence Avoidance

This was taught in stage 1. Quiz the student.

- Explain how wake turbulence is created:
 - Pressure differential → low pressure above and high pressure below the wing.
 - Air moves from the high-pressure area under the wing to the low-pressure area over the wing.
 - It goes around the wing tip and then trails behind.
 - As air curls around the tip, it combines with the wash to form a fast spinning trailing vortex.
 - Right wing tip - counterclockwise.
 - Left wing tip - clockwise.
- Explain how to avoid wake turbulence:
 - Vortices are greatest when an aircraft is in a heavy, slow, and clean configuration.
 - Avoid flying through another aircraft's flightpath.
 - Rotate prior to proceeding aircraft.
 - Avoid following aircraft on similar flight paths.
 - Land past the touchdown point of the other aircraft.
 - Wind is an important factor to wake turbulence.

Stall/Spin Awareness

- Define what an aircraft upset is.
 - "Unusual attitude" or loss of "positive control".
 - Use "Chapter 5 - Maintaining Aircraft Control..." in the AFH.
- Define what a spin is.
- Describe in general terms what a spin is, and the common phase of flight a spin might occur.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Before Takeoff Check

- Verify the use of checklists and correct procedure.
- At this point in flight training, the student must be able to conduct the before takeoff check without instructor guidance.

Emergency Procedure (Electrical)

- Once in the practice area, simulate an electrical failure by announcing that the alternator has failed.
- Verify the student's thought process and understanding of the situation.
- Ask the student to brief what they would do in a situation like such.
 - Troubleshoot.
 - Shed electrical load.
 - Declare an emergency (if required).
 - Land as soon as practical.

Steep Turns

- Instructor does, instructor says:
 - Demonstrate a steep turn.
 - Talk about when to add power, trim, and back pressure.
 - Ensure that the student understands that as the bank is increased, vertical lift decreases, and back pressure must be increased.
 - Emphasize looking outside and maintaining a constant sight picture.
 - Point out which instrument must be referred to.
 - Attitude, airspeed, altimeter, VSI, HSI.
 - Roll out $\frac{1}{2}$ the bank angle before the desired heading.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through when to add power, trim, and back pressure.
 - Coach the student through maintaining a proper sight picture.
 - Where the nose of the airplane or the panel crosses through the horizon.
 - Ensure that the student is looking both outside and inside.
 - Verify the roll out happens at $\frac{1}{2}$ the bank angle before the desired heading.

Short-Field Takeoff and Maximum Performance Climb

- Instructor does, instructor says:
 - Demonstrate a short field takeoff.
 - Talk through runway entry, takeoff roll, and climb out.
 - Emphasize pitching for an attitude and not chasing an airspeed.
 - Use a mix of visual and instrument references.
- Student does, student says:
 - Verify correct procedure.

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- Ensure that the student holds the brakes until full static RPM.
- Verify that the climb out is conducted at 63 knots.
- Verify that the student says obstacle cleared if there is an obstacle.
- Verify that the student says positive rate of climb before retracting the flaps.

Short-Field Approach and Landing

- Instructor does, instructor says:
 - Demonstrate a short field landing.
 - Pick a touchdown point and aiming point on the downwind.
 - Talk through the difference between a normal approach and a short field approach
 - Emphasize trimming!
 - Talk through maintaining a steeper rate of descent and focusing on the aiming point.
 - Talk through the round out and flare.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through staying higher on the approach and slower.
 - Coach the student through maintaining a steeper rate of descent and focusing on the aiming point.
 - Help on the round out and flare if needed.
 - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
 - Verify “maximum” braking.

Soft-Field Takeoff and Climb

- Instructor does, instructor says:
 - Demonstrate a soft field takeoff.
 - Talk through runway entry, takeoff roll, and climb out.
 - Emphasize pitching for an attitude and not chasing an airspeed.
 - Use a mix of visual and instrument references.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through maintaining full aft stabilator and staying in ground effect.
 - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
 - Verify that the student says obstacle cleared if there is an obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.

Soft-Field Approach and Landing

- Instructor does, instructor says:
 - Demonstrate a soft field landing.
 - A normal approach must be conducted.
 - Talk through the round out, staying in ground effect, and timing the flare.
- Student does, student says:
 - Verify correct procedure.

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- Coach the student through the round out, staying in ground effect, and flaring.
- Verify minimal braking.

Forward Slip to Landing

- On one of the traffic patterns, have the student purposely come in high.
- Guide (if needed) the student through conducting a forward slip.
 - Power should be at idle.
 - Pitch for 63 knots.
 - Ailerons into the wind.
 - Rudder opposite.
 - Note: the nose of the airplane will not be aligned with the runway.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 125.
- Assign homework:
 - Review chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short and soft field takeoffs.
 - Review chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short and soft field landings.
 - Review the Warrior's Maneuvers' Manual:
 - Short and soft field takeoffs.
 - Short and soft field landings.

Completion Standards

This lesson is complete when the student is able to:

1. Demonstrate a basic knowledge and understanding of short-field and soft-field takeoffs and landings.
2. Execute short and soft-field takeoffs and landings with instructor guidance.
3. Conduct all normal, abnormal and emergency checklist procedures without instructor guidance.
4. Conduct ground reference maneuvers without instructor guidance and maintain altitude ± 150 ft and airspeed ± 15 kts.